

Harsono Abel Haris

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Education

*Sep. 2021—
Jun. 2020*

Degree: Bachelor of Engineering in Computer Engineering
Where: The University of Hong Kong, Hong Kong
GPA: 3.64 / 4.30
First Class Honours
Dean's Honours List (2024 - 2025)

Research

*September 2024—
May 2025*

Project: Large 3D Reconstruction By Chaining
Where: The University of Hong Kong
Advisor: Dr. Xiaojuan Qi
Contributions:

- Developed an approach for performing 3D reconstruction for large scenes by chaining smaller reconstructions which allows for an efficient and simpler reconstruction processes.

Employment

*January 2024—
October 2025*

Position: Software Engineer
Where: Rabbit Credit Limited, Hong Kong

- Worked in a team of 3 - 4 people to develop a Java spring-boot application for integration with the ICLNET2 system by HKMA that handles sensitive customer data
- Developed an application development pipeline using github actions and docker containers deployed on a windows server to simplify and automate deployment
- Created isolated systems that can communicate with one another by utilizing Docker containers and Nginx allowing for less constraints in development processes.
- Developed a Loan Management System with the PERN stack for a more efficient and customizable loan lending platform

*January 2025—
May 2025*

Position: Student Teaching Assistant - ENGG1330
Where: The University of Hong Kong, Hong Kong

- Hold weekly tutorial session to aid students attending the course to ease them into programming with Python.
- Create tutorial problems weekly for students to solve during tutorial sessions to confirm and challenge their understanding.

*December 2024—
January 2025*

Position: STEM Tutor
Where: Hong Kong Chamber of Speech and Debate, Hong Kong

- Taught middle school students basics of robotics and computing using Arduino
- Held several cases for english debate classes for primary school students

*June 2023—
August 2023*

Position: Automation Engineer Intern
Where: InstantShare Technology Limited, Hong Kong

- Used Slack, Excel Sheet and App Script to develop a simple task management system which allows for more flexibility and interactivity between application testers and developers
- Developed a PoC for a web3 application using Flutter that allows for trading solely within through the app removing the need of going to a different app during a transaction

*January 2023—
May 2023*

Position: Student Teaching Assistant - ENGG1340
Where: The University of Hong Kong, Hong Kong

- Held Q&A sessions for student attending the course

*June 2022—
August 2022*

Position: Quality Assurance Engineer Intern
Where: The University of Hong Kong, Hong Kong

- Worked in a team of around 7 people for running and creating test cases for benchmarking a mobile application used for file sharing across devices of different models

Projects

2026

Package Mapper *springboot, React, D3*

- A visualizer that utilises D3.js to visualize dependencies between files from a github repository's commit history as a graph

2025	Image Scaling Service <i>S3 Bucket, SQS</i> • Deployed an amazon aws node that can accept requests of re-sizing an image.
2025	24 Card Game <i>Java RMI, Java Swing, JDBC, glassfish</i> • Developed a game server and client for playing the 24 card game. The game server contains a play room for 2-4 players and a timeout mechanism. • Used MySQL database using JDBC to store game information and to display game-related information
2024	Learned Step-Size Quantization <i>Pytorch</i> • Implemented an unofficial implementation of the Learned Step Size Quantization method (Steven K. Esser et al. <i>Learned Step Size Quantization</i>. 2020. arXiv: 1902.08153 [cs.LG]. URL: https://arxiv.org/abs/1902.08153)
2024	Simple Networked Gamehouse <i>Python, Multi-threading, Networking</i> • Developed a game server and client that is thread safe utilizing semaphores. The server would allow multiple games in separate rooms to take place at the same time
2024	Contemporary Web Apps <i>HTML, CSS, JS, React, Express, MongoDB</i> • Developed a web application for fetching airport flight information and a drag-and-drop tic-tac-toe game using html, css, and js • Developed a course database viewing application with a token based authentication system using Express and mongoose • Developed an item listing application which allows users to update, upload, and delete items from the list using React. The backend was developed using Express and Mongoose
2023	Music Decoder <i>FPGA</i> • Developed a decoder that can decode a sound byte that contains a message. The decoder consists of 3 main components: Signal decoder, Symbol decoder, and a UART component. The message can be viewed with softwares like putty.
2023	Multi-threaded llama2 <i>C, Multi-threading</i>

- Modified a Large Language Model (LLM) by Andrej Karpathy. In particular, the matrix multiplication part of the code was changed so it can be done in parallel. This results in quicker story generation by the LLM.

2023

Automated Library Cataloging Robot *Raspberry PI, Arduino, RFID*

- A group project which comprises of 4 members. We developed an Automated Guided Vehicle which can follow a certain path to scan books and other related things that have an RFID tag using an RFID scanner. To reach higher levels, the robot has a lift on which the scanners are attached. The robot also has an authentication system using fingerprint detection. The whole robot can be operated with a touch screen.

2023

Web Socket Tic-Tac-Toe *Java*

- A Two player tic-tac-toe game developed using web sockets following a server-client communication model.

Honors and awards

- Entrance Scholarships in Electrical and Electronic Engineering (2023)
- Undergraduate Research Fellowship Programme and Research Internship Award (2024)

Miscellaneous

- *Languages*: English (Proficient), Indonesian (Native)
- *Research Interests*: Distributed Systems, Computer Vision